



Specialized Topics of Theoretical Physics

Majorana Modes in the Machine –
Simulating Topological Phases with Quantum Circuits
and AI

Measuring Topology on Circuits (VQE + String Order)

Zhengyi Liu
[3785549]

Supervisor: Prof. Dr. Bernd Rosenow

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1 Introduction

The one-dimensional Kitaev chain supports boundary Majorana modes in its topological phase [2]. After a Jordan–Wigner transformation, however, neither the Majorana wave functions nor an exact eigenvector are direct hardware observables. The relevant experimental question is therefore more operational: can a parameterized quantum circuit prepare the finite-chain ground state and can measurements of that circuit distinguish the topological and trivial regimes?

This report covers Part 4 of the project, *Measuring Topology on Circuits*. The qubit Hamiltonian and parity convention are taken as inputs from J. Singh’s Part 3 report rather than re-derived here. I construct the ideal circuit workflow: ground-state preparation with a variational quantum eigensolver (VQE), comparison of local and non-local observables, basis-rotated shot readout of the edge string, a chemical-potential sweep, parity-constrained optimization, and an ansatz-depth diagnostic. The observable used below is a finite-size boundary/string diagnostic; it is not, by itself, a replacement for the bulk winding number. Gate, relaxation, and readout-noise effects are outside the present scope and are treated in D. Stoev’s Part 5 report.

2 Variational preparation and circuit measurement

2.1 Hamiltonian, ansatz, and parity constraint

For an even open chain of length L , the encoded Hamiltonian used in Block 3 is

$$H(\mu) = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t - \Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t + \Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (2.1)$$

The numerical sweeps fix $t = \Delta = 1$, so the XX term vanishes and the known transition scale is $|\mu| = 2t$. VQE replaces direct diagonalization by the trial state

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0 \cdots 0\rangle, \quad \boldsymbol{\theta}^*(\mu) = \arg \min_{\boldsymbol{\theta}} C_{\mu}(\boldsymbol{\theta}), \quad (2.2)$$

where U is Qiskit’s hardware-efficient `EfficientSU2` circuit restricted to R_Y rotations and linear nearest-neighbour CNOT entanglement. This is a standard hybrid variational strategy [5, 4, 1]. With r ansatz repetitions, the circuit contains $L(r + 1)$ rotation angles and $r(L - 1)$ CNOTs. For the production sweep, $L = 4$ and $r = 3$, hence there are 16 variational parameters and three entangling layers.

The Hamiltonian conserves the parity operator

$$P = \prod_{j=0}^{L-1} Z_j. \quad (2.3)$$

A low energy alone is insufficient near the nearly degenerate parity sectors, so the Week 7 sweep minimizes

$$C_{\mu}(\boldsymbol{\theta}) = \langle H(\mu) \rangle_{\boldsymbol{\theta}} + \lambda(1 - \langle P \rangle_{\boldsymbol{\theta}})^2, \quad \lambda = 3, \quad (2.4)$$

which targets the even sector. The earlier Week 5 convergence demonstration used the linear bias $\langle H \rangle - 0.1\langle P \rangle$; Eq. (2.4) is the more robust cost used for the full μ continuation.

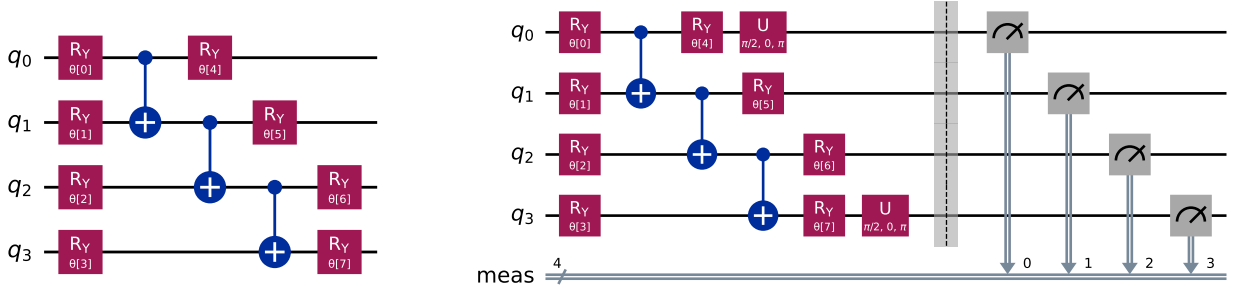


Figure 1: Left: one R_Y -linear-CNOT repetition of the $L = 4$ EfficientSU2 ansatz. Right: the same preparation followed by endpoint basis rotations and computational-basis readout for the four-qubit edge string.

2.2 Why the observable must be non-local

A single endpoint operator such as X_0 anticommutes with parity. In a parity eigenstate $P|\psi\rangle = p|\psi\rangle$,

$$\langle X_0 \rangle = \langle \psi | P X_0 P | \psi \rangle = -\langle X_0 \rangle = 0. \quad (2.5)$$

Thus a local parity-odd edge expectation cannot diagnose the phase. The parity-preserving boundary correlator used in the project is

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1} = i b_0 a_{L-1}, \quad [O_{\text{edge}}, P] = 0, \quad (2.6)$$

where a_j, b_j are the Majorana operators in the project's fixed convention. Only $|\langle O_{\text{edge}} \rangle|$ is plotted, so an overall sign from an alternative Majorana ordering is irrelevant. The string measures coherence between both ends of the finite chain: it is large around the topological sweet spot and suppressed when the end correlation is lost.

2.3 Basis rotation and shot estimator

Quantum hardware returns computational-basis bits. The X factors in Eq. (2.6) are therefore rotated to Z by Hadamard gates on qubits 0 and $L - 1$, while the interior Z factors require no rotation. A measured bit string $b^{(s)} = (b_0^{(s)}, \dots, b_{L-1}^{(s)})$ gives one ± 1 sample

$$m_s = (-1)^{\sum_{j=0}^{L-1} b_j^{(s)}}, \quad \widehat{\langle O_{\text{edge}} \rangle} = \frac{1}{N} \sum_{s=1}^N m_s. \quad (2.7)$$

For a Pauli observable, the sampling standard error is $\sqrt{(1 - \langle O \rangle^2)/N}$ and is at most $N^{-1/2}$. The main sweep uses $N = 4096$, giving a worst-case statistical scale of approximately 0.016.

3 Methods and results

3.1 Numerical protocol and validation

The project's Block 3 implementation uses Qiskit statevector/Aer simulation [3, 6]. ED serves only as a small- L certificate; VQE prepares the states plotted below. Table 1 lists the production settings.

quantity	value
chain and couplings	$L = 4, t = \Delta = 1$
sweep	81 points, $\mu/t \in [-3.5, 3.5]$
ansatz	R_Y + linear CNOT, $r = 3, N_\theta = 16$
optimizer	COBYLA, at most 500 iterations per start
symmetry penalty	$\lambda(1 - \langle P \rangle)^2$, $\lambda = 3$
continuation	previous optimum as the next initial point
recovery	up to 4 additional random restarts
readout	4096 shots for O_{edge}
acceptance	$ \Delta E \leq 0.05$, $ 1 - \langle P \rangle \leq 0.01$, $\mathcal{F} \geq 0.99$

Table 1: Settings of the parity-constrained VQE sweep.

The sweep is warm-started: $\theta^*(\mu_k)$ initializes the optimization at the neighbouring point μ_{k+1} . If a point fails any of the three acceptance tests, random restarts are evaluated. For this validation study the final candidate is selected by its fidelity to the ED even-sector ground state. This ED-assisted post-selection is deliberately stringent for a benchmark, but it is not a device-only selection rule and should not be confused with an experimentally deployable VQE stopping criterion.

3.2 Local versus non-local readout

Figure 2 gives the first operational check. The local parity-odd observable is statistically compatible with zero in all three optimized states, consistent with the parity argument in Eq. (2.5). The non-local string instead separates representative regimes: its magnitude is about 0.99 at $\mu = 0$, approximately 0.24 at the bulk-critical parameter $\mu = 2t$, and about 0.11 at $\mu = 3t$. This representative-point test uses 8192 shots per observable; the full continuation below uses 4096. The result is not merely “measure an edge qubit”: both endpoints and the Jordan–Wigner Z string are essential.

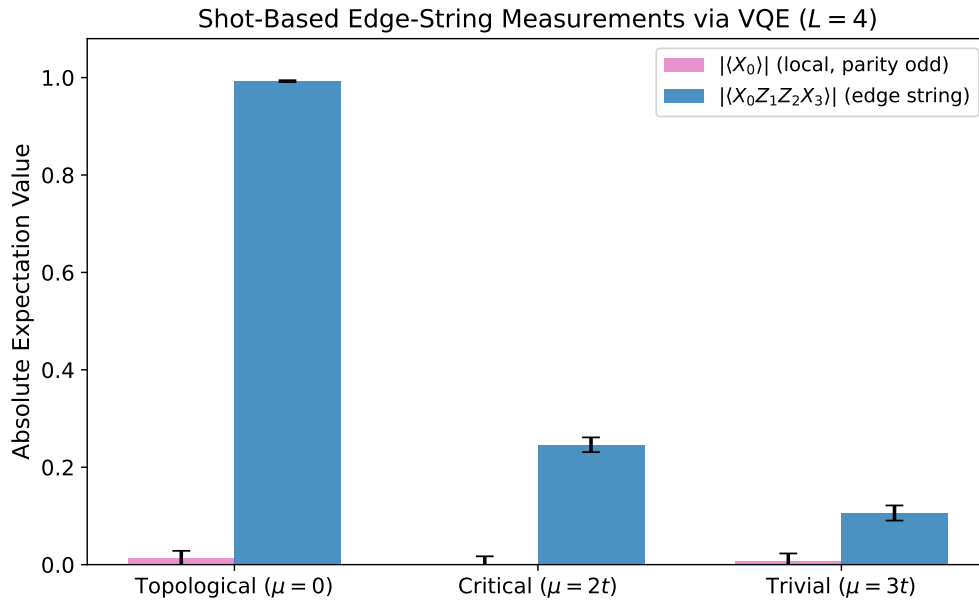


Figure 2: Shot-based VQE measurement of a local parity-odd operator and the non-local edge string for topological, critical, and trivial representative points. Error bars show finite-shot uncertainty.

3.3 Parity-constrained VQE across the phase sweep

The central result is Fig. 3. The ED reference, noiseless VQE expectation, and the 4096-shot estimator lie on the same finite-size crossover. The signal reaches essentially one at the sweet spot $\mu = 0$, falls to about 0.23 at $|\mu| = 2t$, and approaches roughly 0.05 by $|\mu/t| = 3.5$. A four-site system cannot display a discontinuous thermodynamic transition; the dashed lines therefore mark the known bulk boundary while the measured string changes smoothly around it. Shot fluctuations remain consistent with the $N^{-1/2} \simeq 0.016$ worst-case scale and become smaller near unit expectation.

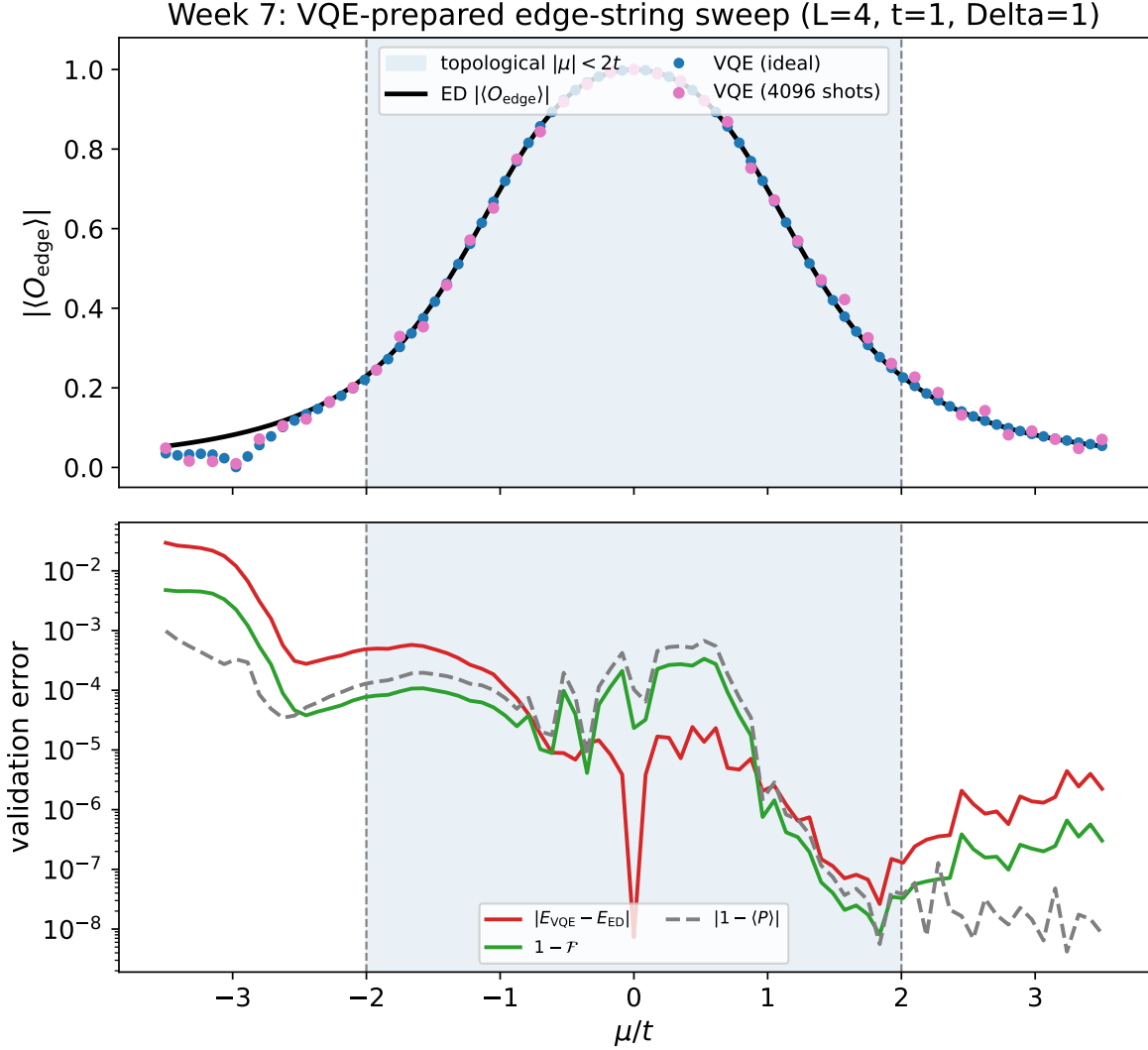


Figure 3: Top: ED, ideal VQE, and finite-shot edge-string curves across the chemical-potential sweep. Bottom: energy error, infidelity, and parity deviation used to validate every prepared state.

The lower panel is important because agreement of one observable alone could be accidental. In the displayed run the energy error stays below 3×10^{-2} , the infidelity below approximately 5×10^{-3} , and the parity deviation below 10^{-3} , all within the acceptance thresholds in Table 1. The larger errors at the first, negative- μ end of the curve are an optimization-history effect: the first point starts randomly, while later points benefit from continuation. They should not be interpreted as a physical $\mu \leftrightarrow -\mu$ asymmetry.

3.4 How deep must the noiseless ansatz be?

The depth scan in Fig. 4 repeats the optimization with six random starts at $\mu/t = 0, 2, 3$ and chooses the ED-validated best state at each depth. One repetition is under-expressive: the infidelity is of order unity in the topological and critical cases and remains just above the 10^{-2} target in the trivial case. At $r = 2$ all three representative points cross below the fidelity threshold; additional repetitions remain accurate but do not improve monotonically because the classical optimization itself is stochastic and non-convex. Choosing $r = 3$ for the full sweep is therefore conservative: it adds one layer beyond the observed noiseless expressibility threshold without making the circuit unnecessarily deep. The separate question of how gate noise changes this choice belongs to Part 5.

Week 7: ansatz-depth diagnostic ($L = 4$, $t = \Delta = 1$)

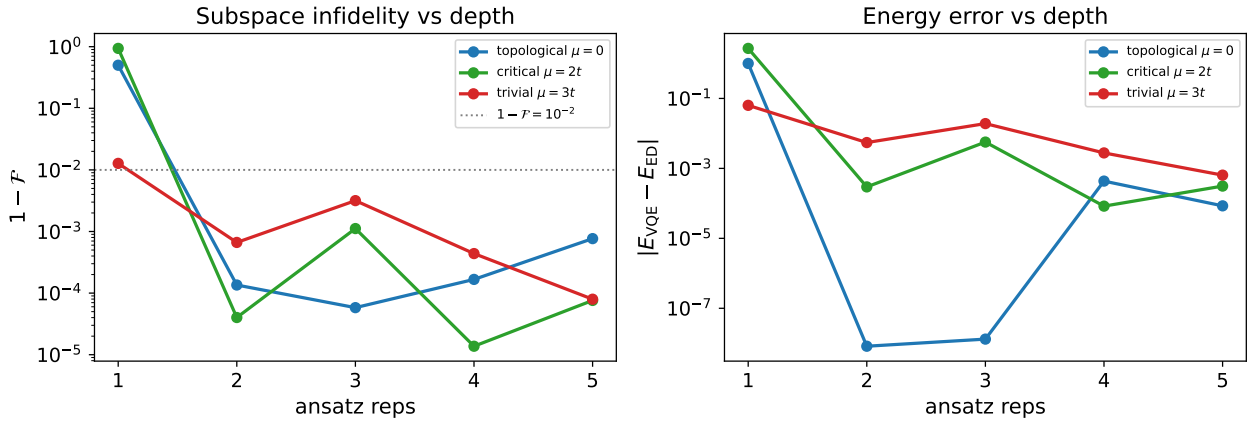


Figure 4: Noiseless ansatz-depth diagnostic at representative topological, critical, and trivial points. Two repetitions are already sufficient for the $L = 4$ fidelity target; $r = 3$ is used in the full sweep.

The ideal Block 3 workflow therefore closes the measurement loop: a gate-based VQE state, symmetry control, basis-rotated bitstring processing, and a non-local observable reproduce the expected finite-size phase diagnostic. Its principal limitations are equally clear: the demonstration uses only $L = 4$, relies on ED for validation and restart ranking, and contains sampling noise but no physical gate or decoherence channel.

4 On the AI collaboration

Large language models were used as an explanatory and auditing tool for this part of the project. They helped organize the VQE-to-measurement workflow, compare the weekly notes with the implementation, and improve the presentation of the parity and shot-estimator arguments. Their suggestions were checked against `block3_core.py`, the generated figures, and exact-diagonalization benchmarks. Two corrections were especially important: the edge string is a finite-size boundary diagnostic rather than a standalone bulk invariant, and the restart ranking in the depth/sweep benchmarks is ED-assisted. The numerical curves themselves come from the project code, not from a language model.

References

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